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WP-2026 SERIES

GLAZOV — CHEPETSK MECHANICAL PLANT

UF_6 Conversion — Zirconium Alloy — Uranium Metal

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

CONVERSION TIER

TARGETABLE

NODE ID	34 — SENI-11
LOCATION	Glazov, Udmurt Republic, Russia
COORDINATES	58.14°N, 52.66°E
OPERATOR	Chepetsk Mechanical Plant (ChMP) / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Conversion Tier
PROCESS ROLE	Mid-Cycle — UF_6 Conversion, Uranium Metal Production, Zirconium Alloy Fabrication
STATUS	TARGETABLE — TVEL SDN constructive coverage; chokepoint function makes this a high-leverage designation target

KEY JUDGMENTS

KJ 1

ChMP is Russia's primary non-ZATO UF_6 conversion facility and the sole domestic producer of zirconium alloys for nuclear fuel rod cladding. Disrupting ChMP would cascade pressure through all four enrichment sites, which cannot operate without UF_6 feedstock.

ALMOST CERTAINLY — **HIGH CONFIDENCE**

KJ 2

ChMP's non-ZATO location in Glazov means it has accessible commercial banking infrastructure — greater financial enforcement leverage than closed-city enrichment sites, and greater front-company procurement exposure.

KJ 3	LIKELY — MODERATE CONFIDENCE TVEL JSC's April 2024 SDN designation constructively covers ChMP. Whether ChMP is explicitly listed requires OFAC confirmation. The chokepoint function — blocking the entire downstream enrichment chain — makes explicit listing a higher priority here than at downstream fabrication nodes. LIKELY — MODERATE CONFIDENCE

I. FACILITY OVERVIEW

ATTRIBUTION	TVEL JSC annual reporting and the NTI Russia Nuclear Profile identify ChMP at Glazov as the primary conversion facility routing U_3O_8 yellowcake to UF_6 for the Russian enrichment network, and as the sole domestic producer of zirconium alloys used for fuel rod cladding [TIER 1 — TVEL Annual Report; TIER 2 — NTI Russia Profile 2023].
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that ChMP is a dual chokepoint — simultaneously the conversion gateway to all four enrichment sites and the zirconium supply source for both fabrication facilities. Enforcement action at ChMP has cascading effect on a broader range of downstream nodes than any single enrichment site designation.
LIMITS	The public record does not establish ChMP's specific annual UF_6 output volume, the proportion of zirconium alloy production destined for export versus domestic use, or the current status of uranium metal production at the facility.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
U_3O_8 feedstock input	Receives yellowcake from Krasnokamensk (Node 27) and other ARMZ mining operations
UF_6 conversion output	→ Enrichment at Novouralsk (Node 28), Zelenogorsk (Node 30), Seversk (Node 29), Angarsk (Node 31)
Zirconium alloy output	→ Fuel fabrication at Elektrostal (Node 32) and Novosibirsk NCCP (Node 33) for fuel rod cladding
Uranium metal output	→ Rosatom research programs; potential export under civilian contracts

III. FINANCIAL NETWORK

ATTRIBUTION	ChMP's financial flows route through TVEL JSC, designated by OFAC on April 19, 2024 [TIER 1 — OFAC SDN List]. As a non-ZATO facility, ChMP has commercial banking relationships more accessible than closed-city sites. Zirconium alloy international sales generate commercially visible trade flows partially trackable through customs data.
JUDGMENT	Available reporting supports a moderate-confidence judgment that ChMP's non-ZATO commercial banking infrastructure represents a more accessible financial enforcement target than the ZATO enrichment facilities. Zirconium alloy export controls provide a separate BIS enforcement avenue independent of OFAC SDN action.
LIMITS	The public record does not establish the specific Western banks currently clearing ChMP commercial transactions, specific zirconium alloy export customers, or uranium metal output volumes and customers.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	ChMP designatable as TVEL/Rosatom subsidiary. UF_6 conversion is direct proliferation-enabling activity.
CAATSA §231	Defense/Intel Sector	Rosatom sector designation covers ChMP.
IEEPA / OFAC	Entity Designation	Explicit SDN listing recommended. TVEL designation constructive coverage status unconfirmed.
EAR / BIS	Nuclear Referral / ECCN	Zirconium alloy exports controlled under nuclear-related materials; BIS coordination applicable.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024, constructively covering ChMP as a TVEL subsidiary [TIER 1 — OFAC SDN List]. ChMP is not explicitly listed on the SDN list as of March 2026.
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that ChMP explicit SDN listing is the highest-leverage single designation action available in the conversion tier. The chokepoint logic runs upstream: disrupting UF_6 conversion

	directly affects the weapons-relevant enrichment step, not merely the civilian fuel delivery endpoint.
LIMITS	The public record does not establish the lead time between UF_6 conversion disruption and enrichment cascade operational impact, or the extent of UF_6 inventory stockpiles at enrichment sites that would buffer against conversion disruption.

VI. ENFORCEMENT GAPS

Gap 1 — Explicit SDN Listing Unconfirmed

ChMP's coverage under TVEL's April 2024 SDN designation is ambiguous. Explicit listing would close this gap and make ChMP-specific financial pressure actionable for U.S. persons and third-country compliance purposes.

Gap 2 — Zirconium Export Monitoring

Commercial zirconium alloy exports from ChMP are not comprehensively tracked in open-source trade data. End-use monitoring for nuclear-grade zirconium sales to non-allied customers is the primary surveillance gap at this node.

Gap 3 — Uranium Metal Production Accountability

Uranium metal output from ChMP does not receive equivalent IAEA safeguards attention as UF_6 conversion. Uranium metal is a weapons-relevant material form warranting independent material accountancy. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on known IAEA safeguards focus on UF_6 conversion rather than uranium metal.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. An OFAC action to verify and pursue explicit SDN listing of ChMP under EO 13382 as a TVEL subsidiary and independent UF_6 conversion facility.
2. Implementation of enhanced end-use monitoring for zirconium alloy exports from ChMP to all non-allied customers.
3. IAEA should be engaged on enhanced safeguards attention to uranium metal production at ChMP — specifically material accountancy for non-UF_6 uranium outputs.
4. Modeling of the fuel cycle disruption impact of ChMP enforcement action: quantify lead time between UF_6 conversion disruption and enrichment cascade operational impact across all four enrichment sites.

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	Named Western counterparty not identified for zirconium alloy exports. Upgrade: Eurostat COMEXT HS 8109 (zirconium), Russia origin, 2019–2024. Expected: Czech CEZ/Dukovany, Finnish

		Fortum historical, Hungarian MVM as likely named counterparties.
2	THRESHOLD	GL 115A expiry June 30, 2025 is the primary enforcement trigger. Post-expiry, EU banks clearing ChMP payments face EO 14024/14114 secondary sanctions exposure. Monitor OFAC for successor license issuance.

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WP-2026 SERIES

ZABAYKALSK RAIL TERMINAL

Russia-China Rail Border Crossing — Trans-Siberian Eastern Terminus

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

TRANSPORT TIER

TARGETABLE

NODE ID	35 — SENI-12
LOCATION	Zabaykalsk, Zabaykalsky Krai, Russia
COORDINATES	49.65°N, 117.33°E
OPERATOR	Russian Railways (RZhD) / Federal Customs Service (FCS)
NETWORK GROUP	Russian Nuclear Fuel Cycle — Transport / Logistics Tier
PROCESS ROLE	Transport Chokepoint — Primary Russia-China Rail Border Crossing for Eastern Fuel Cycle
STATUS	TARGETABLE — logistics intermediary designation available; direct U.S. enforcement jurisdiction limited

KEY JUDGMENTS

KJ 1

Zabaykalsk is the primary Russia-China rail border crossing, approximately 70 km from Krasnokamensk (PIMCU, Node 27). The gauge-change dwell creates a mandatory physical interdiction window for eastbound cargo. All declared nuclear material moving overland from the eastern fuel cycle to China transits this crossing.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Once nuclear material crosses at Zabaykalsk, IAEA safeguards continuity depends entirely on Chinese declarations and inspections. No independent verification mechanism for material accountability transitions exists at the Russia-China material

KJ 3	transfer point. LIKELY — MODERATE CONFIDENCE The logistics intermediary chain connecting Rusatom Service to Chinese customs brokers and freight forwarders at Zabaykalsk has not been documented in open-source analysis. This is the primary financial intelligence gap at this node. LIKELY — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED

I. NODE OVERVIEW

ATTRIBUTION	Russian Railways (RZhD) public infrastructure documentation identifies Zabaykalsk as the primary Trans-Siberian main line border crossing between Russia and China in Zabaykalsky Krai [TIER 2 — RZhD infrastructure reports; TIER 3 — Railway Gazette International]. The crossing involves gauge-change operations between Russian broad gauge (1520mm) and Chinese standard gauge (1435mm), creating a mandatory cargo dwell at the border facility.
JUDGMENT	Available reporting supports a high-confidence judgment that Zabaykalsk is the physical chokepoint for nuclear material moving overland from the eastern Russian fuel cycle to China. The gauge-change dwell creates the primary physical interdiction opportunity in this logistics corridor.
LIMITS	The public record does not establish whether Chinese customs at the Manzhouli receiving terminal operates radiation detection equipment adequate to identify anomalous nuclear material in commercial freight, or whether any manifest verification mechanism for nuclear material declarations exists at the crossing.

II. LOGISTICS CONNECTIONS

CONNECTION	DETAIL
Upstream origin	Krasnokamensk (PIMCU, Node 27) → Trans-Siberian rail → Zabaykalsk crossing
Downstream China	→ Manzhouli (China) → Chinese rail network → CNNC fuel cycle facilities
Parallel crossing	Manzhouli road crossing (~30 km north) — secondary route for road transport
Rusatom Service	Rosatom's logistics subsidiary manages nuclear material transport documentation within Russia

III. FINANCIAL NETWORK AND ENFORCEMENT LEVERS

ATTRIBUTION	Rusatom Service — Rosatom's logistics subsidiary — manages nuclear material transport documentation for fuel cycle shipments within Russia. RZhD generates freight billing for cargo transiting Zabaykalsk. Chinese customs brokers and freight forwarders operating at Manzhouli constitute the financial interface on the Chinese side of the crossing.
JUDGMENT	Available reporting supports a low-moderate-confidence judgment that the freight forwarder and customs broker chain between Rusatom Service and Chinese fuel cycle facilities represents a traceable financial network with potential Western banking exposure. This chain has not been mapped in open-source analysis.
LIMITS	The public record does not establish which specific entities handle nuclear material logistics at Zabaykalsk and Manzhouli, or whether any of these entities have Western financial relationships accessible to OFAC enforcement.

IMPLEMENTATION FRICTION Direct U.S. enforcement at Zabaykalsk is not available — it is a Russian border facility outside U.S. jurisdictional reach. Enforcement instruments are indirect: secondary sanctions on logistics intermediaries, diplomatic pressure on Chinese customs standards, and IAEA safeguards continuity engagement.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Transport Logistics	Entities providing transport, freight forwarding, or customs services for designated Rosatom entities are exposed to secondary designation.
IEEPA / OFAC	Logistics Intermediary	Freight forwarders and customs brokers facilitating nuclear material transport for SDN-listed entities are designatable.
CBP / DOE	Radiation Detection	U.S. cannot inspect Russian-Chinese border crossings directly; diplomatic pressure on Chinese customs radiation monitoring standards is available.
IAEA	Safeguards Continuity	IAEA safeguards continuity verification at the Russia-China material transfer point available through IAEA Board of Governors.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC has designated entities providing logistics support to SDN-listed Russian entities under EO 13382 in other sanctions contexts [TIER 1 — OFAC enforcement actions 2022–2024]. Rusatom Service is a Rosatom subsidiary providing nuclear material logistics; TVEL JSC's April 2024 SDN designation constructively covers Rusatom Service.
JUDGMENT	Available reporting supports a moderate-confidence judgment that the most actionable enforcement instruments at Zabaykalsk are secondary sanctions targeting logistics intermediaries on the Chinese side of the crossing with Western financial exposure, combined with diplomatic engagement through IAEA on safeguards continuity at the material transfer point.
LIMITS	The public record does not establish whether any Chinese logistics entities at Manzhouli have Western financial relationships that would be accessible to OFAC secondary sanctions enforcement.

VI. ENFORCEMENT GAPS

Gap 1 — Logistics Intermediary Chain Unmapped

The freight forwarder and customs broker chain connecting Rusatom Service to Chinese fuel cycle facilities via Zabaykalsk has not been documented in open-source analysis. Mapping this chain is the prerequisite for any secondary sanctions action at this node.

Gap 2 — Radiation Detection Standard at Crossing Unknown

Whether Chinese customs at Manzhouli operates radiation detection equipment capable of identifying anomalous nuclear material is not confirmed in open-source reporting. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED that standard commercial customs screening is inadequate for nuclear material verification.

Gap 3 — IAEA Safeguards Material Transfer Verification

No independent mechanism exists for verifying that declared nuclear material manifests at Zabaykalsk accurately reflect shipment contents. IAEA safeguards operate on declared quantities; undeclared material would not be captured at the crossing point.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Financial intelligence mapping of Rusatom Service's logistics contractor and freight forwarder relationships for the Krasnokamensk-to-China corridor; identify entities with Western financial exposure for secondary sanctions targeting.
2. Engagement through Chinese counterparts through IAEA Board of Governors on safeguards continuity verification at the Zabaykalsk/Manzhouli material transfer point.
3. A coordinated State-DOE effort to assess radiation detection capability at Manzhouli customs through allied intelligence and diplomatic channels; develop diplomatic pressure track if capability is inadequate.
4. Profiling of the parallel Manzhouli road crossing as a supplementary SENI transport node.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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NAKHODKA / VOSTOCHNY PORT COMPLEX

Pacific Maritime Export Terminal — Russia-DPRK Corridor

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

TRANSPORT TIER

TARGETABLE

NODE ID	36 — SENI-13
LOCATION	Nakhodka, Primorsky Krai, Russia
COORDINATES	42.81°N, 132.88°E
OPERATOR	FESCO / Port Authority of Vostochny / Rosatom logistics chain
NETWORK GROUP	Russian Nuclear Fuel Cycle — Transport / Logistics Tier
PROCESS ROLE	Transport Node — Pacific Maritime Export; Russia-DPRK Corridor Departure Point
STATUS	TARGETABLE — FESCO and vessel operator designation available; DPRK nexus elevates enforcement priority

KEY JUDGMENTS

KJ 1 The Nakhodka/Vostochny port complex is Russia's primary Pacific maritime export terminal and the documented departure point for Russia-DPRK cargo that expanded significantly post-2022. UN Panel of Experts reporting on DPRK sanctions evasion documents use of this corridor for prohibited cargo [TIER 1 — UN Panel of Experts, UNSC 1718 Committee, 2023–2024 reports].

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2 FESCO (Far Eastern Shipping Company) — the primary Russian Pacific carrier — has been identified in open-source sanctions enforcement reporting as operating in

KJ 3	sanctionable corridors post-2022. FESCO and associated vessel operators are candidate targets for OFAC shipping sanctions. **LIKELY — MODERATE CONFIDENCE** The Russia-DPRK maritime corridor presents the most plausible logistics pathway for any small-quantity nuclear material or technology transfer from Russian fuel cycle nodes to DPRK. This assessment does not establish that such transfer has occurred. LOW CONFIDENCE — ANALYTICALLY INFERRED. **ROUGHLY EVEN ODDS — LOW CONFIDENCE — ANALYTICALLY INFERRED**

I. NODE OVERVIEW

ATTRIBUTION	UN Panel of Experts reports to the UNSC 1718 Committee (2023, 2024) document use of the Vostochny/Nakhodka port complex as a departure point for Russia-DPRK cargo including prohibited items under UNSC sanctions [TIER 1 — UN Panel of Experts S/2024/215]. The UN Panel identifies specific vessels and flag registries used in this corridor.
JUDGMENT	Official UN reporting supports a high-confidence judgment that the Vostochny/Nakhodka complex is an active departure point for DPRK-bound cargo that violates UNSC 1718 sanctions. Post-2022 Russia-DPRK strategic alignment has expanded this corridor's operational volume.
LIMITS	The public record does not establish that nuclear material or technology has transited the Vostochny-DPRK corridor. The DPRK proliferation concern at this node is a logistics exposure, not a documented transfer. The available reporting does not permit a firm conclusion on the specific vessels currently operating nuclear-relevant cargo in this corridor.

II. LOGISTICS CONNECTIONS

CONNECTION	DETAIL
Trans-Siberian logistics input	Trans-Siberian rail → Vostochny → maritime routes to Japan, South Korea, China, Northeast Asia
Russia-DPRK corridor	Vostochny/Nakhodka → Chongjin or Nampo (DPRK) — documented post-2022 sanctions-evasion route
FESCO operations	Primary Russian Pacific carrier; documented use of Vostochny for sanctionable cargo
Nuclear fuel export	Sea-routed LEU fuel assemblies to Asian VVER operators as

	alternative to overland routing

III. FINANCIAL NETWORK

ATTRIBUTION	FESCO has historically used Singapore, Hong Kong, and other Asian financial centers for freight clearing. UN Panel of Experts reports identify specific vessel operators and financial intermediaries associated with the Vostochny-DPRK corridor [TIER 1 — UN Panel of Experts reports 2023–2024].
JUDGMENT	Official UN reporting supports a moderate-confidence judgment that FESCO and associated vessel operators have Western financial exposure through Asian financial center correspondent relationships that are accessible to OFAC secondary sanctions enforcement.
LIMITS	The public record does not establish the specific current correspondent banks clearing FESCO's Vostochny-corridor freight revenue, or the extent to which FESCO has migrated to non-Western clearing post-2022.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13722	DPRK Sanctions	Vessels and operators involved in Russia-DPRK cargo via Vostochny exposed to DPRK-nexus designation if carrying prohibited cargo.
EO 13382	WMD Proliferators — Maritime	Maritime transport entities facilitating nuclear material movement for designated Rosatom entities are exposed.
OFAC / GLOMAG	Shipping / Port Designation	FESCO and associated vessel operators are candidate targets under existing Russia and DPRK shipping sanctions authorities.
UNSC 1718	Panel Reporting	UN Panel reporting creates evidentiary record; Russia vetoes enforcement but documentation supports bilateral sanctions.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC has designated maritime shipping entities under EO 13722 and EO 13382 in multiple prior enforcement actions [TIER 1 — OFAC SDN enforcement actions 2018–

	2024]. UN Panel of Experts reports identify specific FESCO-adjacent vessel operators in the Vostochny-DPRK corridor [TIER 1 — UN Panel S/2024/215].
JUDGMENT	Official reporting supports a moderate-confidence judgment that FESCO and vessel operators in the Vostochny-DPRK corridor are designatable under EO 13722 DPRK sanctions authority based on documented cargo in the corridor, independent of any nuclear material nexus.
LIMITS	The public record does not establish that FESCO has directly transported nuclear material or technology to DPRK. The enforcement predicate is the documented DPRK prohibited cargo corridor, not a confirmed nuclear transfer.

VI. ENFORCEMENT GAPS

Gap 1 — FESCO Sanctions Compliance Incomplete

FESCO and associated vessel operators have continued operating in sanctionable corridors despite Western sanctions pressure. Secondary sanctions enforcement against FESCO's Asian financial clearing relationships has not been fully executed.

Gap 2 — Vessel Identification Gap

The specific vessels operating the Russia-DPRK maritime corridor out of Vostochny are not comprehensively identified in open-source reporting. UN Panel reporting covers some operators but is not comprehensive. MODERATE CONFIDENCE based on UN Panel Tier 1 reporting that covers only a subset of active vessels.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Development of FESCO and associated vessel operator designation packages under EO 13722 DPRK sanctions and EO 13382 WMD proliferator logistics authorities.
2. Implementation of port-of-call restrictions in allied Pacific jurisdictions (Japan, South Korea, Australia) for vessels with documented Vostochny-DPRK corridor calls.
3. Coordination through with UN Panel of Experts on DPRK sanctions to develop comprehensive vessel identification for the Vostochny-DPRK maritime corridor.
4. Engagement through IAEA on maritime safeguards continuity verification for nuclear material shipments departing Russian Pacific ports.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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TAYSHET RAIL JUNCTION

Trans-Siberian / BAM Railway Junction — Eastern Fuel Cycle Alternative Routing

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

TRANSPORT TIER

TARGETABLE

NODE ID	37 — SENI-14
LOCATION	Tayshet, Irkutsk Oblast, Russia
COORDINATES	55.93°N, 98.00°E
OPERATOR	Russian Railways (RZhD) — Tayshet Division
NETWORK GROUP	Russian Nuclear Fuel Cycle — Transport / Logistics Tier
PROCESS ROLE	Transport Node — Trans-Siberian / BAM Junction; Alternative Routing for Eastern Fuel Cycle
STATUS	TARGETABLE — RZhD designation assessment warranted; BAM routing for anomalous shipments is ANALYTICALLY INFERRED

KEY JUDGMENTS

KJ 1

Tayshet is the junction point where the Trans-Siberian Railway and the Baikal-Amur Mainline (BAM) diverge. Trans-Siberian routing continues toward Zabaykalsk (China crossing, Node 35). BAM routing branches northeast toward the Russian Far East and DPRK-proximate Pacific ports.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

BAM routing from Tayshet would allow eastbound nuclear-relevant cargo to bypass Trans-Siberian main line monitoring points and reach Russian Far East ports without transiting the Zabaykalsk or Manzhouli crossings. This creates a surveillance gap for

KJ 3	anomalous material movement. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED. **ROUGHLY EVEN ODDS — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED** RZhD is the logistics backbone for all Russian nuclear material transport. RZhD designation under EO 13382 WMD proliferator logistics authority is legally available but would carry major collateral economic effects requiring cost-benefit assessment before execution. **LIKELY (DESIGNATION AVAILABLE) — MODERATE CONFIDENCE**

I. NODE OVERVIEW

ATTRIBUTION	Russian Railways infrastructure documentation identifies Tayshet as the BAM-Trans-Siberian junction in Irkutsk Oblast [TIER 3 — RZhD network maps; Railway Gazette International]. The Trans-Siberian continues east toward Irkutsk, Angarsk (Node 31), and Zabaykalsk. The BAM branches northeast toward Bratsk, Tynda, Komsomolsk-on-Amur, and Russian Far East Pacific ports.
JUDGMENT	Available reporting supports a high-confidence judgment that Tayshet is the routing choice point for eastbound cargo between the southern Trans-Siberian route (declared civilian fuel cycle path toward China) and the northern BAM route (alternative path toward Russian Far East and DPRK-proximate ports). Monitoring cargo manifest routing decisions at Tayshet would provide intelligence on anomalous material movement.
LIMITS	The public record does not establish that nuclear material has been routed via BAM rather than Trans-Siberian. The BAM routing concern is ANALYTICALLY INFERRED from the junction's geographic position and post-2022 Russia-DPRK logistics expansion. The available reporting does not permit a firm conclusion on current cargo monitoring capability at Tayshet junction.

II. LOGISTICS CONNECTIONS

CONNECTION	DETAIL
Trans-Siberian westbound	→ Novosibirsk, Novouralsk, European Russia fuel cycle nodes
Trans-Siberian eastbound	→ Irkutsk, Angarsk (AECC, Node 31), Ulan-Ude, Zabaykalsk (Node 35, China crossing)
BAM northeast branch	→ Bratsk, Tynda, Komsomolsk-on-Amur, Russian Far East Pacific ports

BAM Far East terminus	Vanino / Sovetskaya Gavan ports — secondary maritime export points north of Nakhodka

III. FINANCIAL NETWORK

ATTRIBUTION	Nuclear material freight billing in Russia routes through Rusatom Service rather than standard commercial RZhD channels. RZhD generates standard freight documentation for all cargo transiting Tayshet junction regardless of material type.
JUDGMENT	Available reporting supports a low-moderate-confidence judgment that RZhD freight manifest data for BAM-routed nuclear-relevant cargo would constitute the primary intelligence product available at this node. Allied intelligence sharing on RZhD manifest data is the operative surveillance instrument.
LIMITS	The public record does not establish RZhD's specific freight documentation procedures for nuclear material, or whether BAM-routed nuclear material would generate manifest data distinguishable from civilian freight in available monitoring channels.

IMPLEMENTATION FRICTION Direct enforcement at Tayshet is not available — it is a Russian rail facility outside U.S. jurisdictional reach. The enforcement approach is intelligence-driven: identify nuclear-relevant cargo manifests routing onto the BAM branch rather than continuing on the Trans-Siberian, and use that manifest data to support designation actions against logistics intermediaries.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Transport Logistics	Entities providing BAM routing logistics for nuclear material outside declared fuel cycle channels are exposed to designation.
IEEPA / OFAC	Logistics Intermediary	RZhD designation legally available under WMD proliferator logistics authority. Collateral effects require cost-benefit assessment.
UNSC 1718	DPRK Cargo Monitoring	BAM-routed cargo destined for DPRK-proximate Pacific ports falls within UNSC 1718 cargo monitoring obligations.
Allied Intelligence	Rail Manifest Data	Allied intelligence sharing on RZhD manifest data for BAM-routed cargo is the primary surveillance instrument.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC has designated transportation entities providing logistics support to WMD proliferators under EO 13382 in prior enforcement actions [TIER 1 — OFAC SDN enforcement actions 2022–2024]. RZhD is not currently designated.
JUDGMENT	Official enforcement precedent supports a moderate-confidence judgment that RZhD is legally designatable under EO 13382 WMD proliferator logistics authority. The enforcement question is collateral cost: RZhD designation would be a major economic action affecting all Russian rail commerce, not only nuclear logistics. A full cost-benefit analysis warrants attention before execution.
LIMITS	The public record does not establish the specific BAM cargo volumes carrying nuclear-relevant material, or whether intelligence-driven manifest monitoring at Tayshet is operationally feasible through currently available allied channels.

VI. ENFORCEMENT GAPS

Gap 1 — BAM Cargo Manifest Monitoring Absent

No established mechanism exists for monitoring nuclear-relevant cargo manifests at Tayshet junction for anomalous BAM routing. This is assessed as the primary surveillance gap in the eastern fuel cycle logistics chain. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on known limitations of open-source rail manifest data.

Gap 2 — Vanino / Sovetskaya Gavan Ports Undocumented

The BAM terminus ports at Vanino and Sovetskaya Gavan are not currently documented in the SENI index. They represent secondary maritime export points that could be used to avoid monitoring at Nakhodka/Vostochny (Node 36). Supplementary profiles for these ports are recommended.

Gap 3 — RZhD Designation Collateral Assessment Not Complete

RZhD has not been designated despite functioning as the logistics backbone for all Russian nuclear material transport. The collateral economic effects of RZhD designation have not been publicly assessed. This assessment warrants attention before the designation can be recommended.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. DOE and Development of a full cost-benefit assessment of RZhD designation under EO 13382 WMD proliferator logistics authority, including collateral economic effects across all Russian rail commerce.
2. Commissioning of supplementary SENI profiles for Vanino and Sovetskaya Gavan ports as the BAM maritime terminus nodes.
3. Development of allied intelligence-sharing request for RZhD manifest data on BAM-routed nuclear-relevant cargo from Tayshet junction.
4. Coordination through with UN Panel of Experts on DPRK sanctions regarding BAM-routed cargo monitoring obligations under UNSC 1718.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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SAROV — RFNC-VNIIEF

Nuclear Weapons Design Laboratory — Russian Federal Nuclear Center

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WEAPONS TIER

CONSTRAINED

NODE ID	38 — SENI-15
LOCATION	Sarov, Nizhny Novgorod Oblast, Russia (ZATO)
COORDINATES	54.93°N, 43.32°E
OPERATOR	RFNC-VNIIEF / Rosatom State Atomic Energy Corporation
NETWORK GROUP	Russian Nuclear Weapons Complex — Design Tier
PROCESS ROLE	Weapons — Nuclear Warhead Design, HEU/Pu Specification, Weapons Physics Research
STATUS	CONSTRAINED — no Western financial exposure; BIS Entity List and export controls are the operative instruments

KEY JUDGMENTS

KJ 1 RFNC-VNIIEF at Sarov is Russia's primary nuclear weapons design laboratory, the functional equivalent of Los Alamos National Laboratory. It defines the material specifications — enrichment levels, component geometries — that flow upstream to the enrichment and fabrication facilities.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2 The termination of U.S.-Russia nuclear security cooperative programs post-2022 has eliminated all non-intelligence transparency mechanisms for VNIIEF activities. No arms control verification mechanism currently covers Sarov's weapons design

	programs. ALMOST CERTAINLY — HIGH CONFIDENCE
KJ 3	VNIIEF's dual-use technology procurement through UAE, Turkey, China, and India intermediaries is ongoing. The public record does not permit a firm conclusion on the specific entities involved or the categories of equipment being procured. LIKELY — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED
KJ 4	Potential Russian weapons design technology transfer to DPRK through post-2022 strategic alignment is analytically plausible. This assessment does not establish that such transfer has occurred. LOW CONFIDENCE — ANALYTICALLY INFERRED. Corroboration warrants attention before this can serve as a published enforcement predicate. ROUGHLY EVEN ODDS — LOW CONFIDENCE — ANALYTICALLY INFERRED

I. FACILITY OVERVIEW

ATTRIBUTION	ODNI's Annual Threat Assessment and U.S. State Department arms control compliance reports identify RFNC-VNIIEF at Sarov as Russia's primary nuclear weapons design center [TIER 1 — ODNI ATA 2024; State Department Compliance Report 2023]. The facility is the successor to the Soviet Arzamas-16 complex and has participated in joint research programs with U.S. national laboratories under now-suspended cooperative agreements.
JUDGMENT	Official reporting supports a high-confidence judgment that VNIIEF is the demand-side node for HEU and weapons-grade materials produced at the enrichment tier. It defines the material requirements that make enrichment site diversion scenarios operationally significant.
LIMITS	The public record does not establish the current scope of VNIIEF weapons design activities, specific HEU specifications for current weapons programs, or whether any technology transfer to DPRK has occurred post-2022.

DOCUMENTED STRUCTURAL CONSTRAINT VNIIEF operates entirely within Russia's closed federal nuclear weapons budget. Standard OFAC SDN designation instruments targeting commercial financial relationships have no direct leverage at this node. Monitoring, not financial pressure, is the operative enforcement gap.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
Weapons specifications →	HEU material requirements → UECC (Node 28), SCC (Node 29),

Enrichment	ECP (Node 30), AECC (Node 31)
Weapons specifications → Assembly	Component requirements → Trekhgorny (Node 39), Lesnoy (Node 40)
Weapons specifications → Pu pathway	Pu component requirements → Mayak (Node 41), Zheleznogorsk (Node 42)
Procurement channels	Dual-use technology procurement via UAE, Turkey, China, India intermediaries — ANALYTICALLY INFERRED
Historical cooperation	U.S. national laboratory programs (Los Alamos, Livermore) — now terminated (post-2022)

III. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	ODNI and State Department reporting identify VNIIEF's weapons design function. BIS maintains the Entity List authority to restrict dual-use technology exports to specific entities [TIER 1 — EAR Part 744]. OFAC has designated WMD design facilities under EO 13382 in other national contexts.
JUDGMENT	Official reporting supports a moderate-confidence judgment that BIS Entity List designation is more appropriate than OFAC SDN designation at VNIIEF: it targets the technology transfer vector — Western scientific equipment and software reaching VNIIEF through third-country procurement channels — without imposing the full financial blocking effect of SDN designation, which would have no practical impact given VNIIEF's closed budget.
LIMITS	The public record does not establish which specific dual-use items VNIIEF is procuring through third-country intermediaries, or which intermediary entities in UAE, Turkey, China, and India are involved.

IV. ENFORCEMENT GAPS

Gap 1 — Post-2022 Monitoring Void

Termination of U.S.-Russia nuclear security cooperation has eliminated all non-intelligence transparency mechanisms for VNIIEF activities. No arms control verification mechanism currently covers Sarov's weapons design programs. HIGH CONFIDENCE based on the documented termination of cooperative programs.

Gap 2 — Third-Country Procurement Channels Active

VNIIEF's dual-use technology procurement through UAE, Turkey, China, and India intermediaries is ongoing but not comprehensively mapped. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on known Russian dual-use procurement patterns in these jurisdictions post-2022.

Gap 3 — DPRK Technology Transfer Assessment

Potential weapons design technology transfer to DPRK is LOW CONFIDENCE — ANALYTICALLY INFERRED. This assertion requires corroboration from Tier 1 or credible Tier 2 sources before publication. It should not appear as a stated fact in any version of this profile submitted for peer review.

V. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Development of a VNIIEF Entity List designation package targeting dual-use technology procurement channels; coordinate with allied export control authorities (Wassenaar, NSG members).
2. Financial intelligence mapping of third-country intermediary networks procuring Western dual-use technology for VNIIEF through UAE, Turkey, China, and India channels.
3. Assessment of policy options for re-establishing limited technical monitoring at Sarov through arms control treaty mechanisms or track-2 scientific engagement.
4. IC-connected analysis to develop indicators for Russia-DPRK weapons design information exchange to provide corroboration for the LOW CONFIDENCE transfer assessment before it can serve as a published enforcement predicate.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

TREKHGORNY — PRIBOROSTROITELNY ZAVOD

Nuclear Warhead Component Assembly — Closed City (Zlatoust-36)

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

WEAPONS TIER

CONSTRAINED

NODE ID	39 — SENI-16
LOCATION	Trekhgorny, Chelyabinsk Oblast, Russia (ZATO)
COORDINATES	54.82°N, 58.45°E
OPERATOR	Priborostroitelny Zavod / Rosatom
NETWORK GROUP	Russian Nuclear Weapons Complex — Assembly Tier
PROCESS ROLE	Weapons — Nuclear Warhead Component Fabrication and Assembly
STATUS	CONSTRAINED — no Western financial exposure; export controls on precision manufacturing equipment are operative instrument

KEY JUDGMENTS

KJ 1

Trekhgorny hosts Russia's primary nuclear warhead component assembly facility, approximately 120 km from Novouralsk (UECC, Node 28). This geographic proximity places the primary warhead assembly facility within direct logistics range of Russia's largest enrichment site, minimizing transport exposure of weapons-grade material.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Trekhgorny operates entirely within Russia's classified nuclear weapons program with no commercial activities or Western financial exposure. Standard OFAC SDN designation has no direct leverage. Export controls on precision manufacturing equipment are the operative enforcement instrument.

ALMOST CERTAINLY — HIGH CONFIDENCE

I. FACILITY OVERVIEW

ATTRIBUTION	The Nuclear Threat Initiative's Russia Facilities Profile and U.S. State Department arms control compliance reports identify Trekhgorny — formerly Zlatoust-36 — as a ZATO closed city hosting nuclear warhead component assembly operations [TIER 2 — NTI Russia Facilities; TIER 1 — State Department Compliance Report 2023].
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that Trekhgorny is a warhead assembly endpoint for HEU produced at Novouralsk. The Novouralsk-Trekhgorny logistics corridor — approximately 120 km — is the shortest HEU-to-assembly route in the Russian weapons complex and represents an unmonitored material accountability gap.
LIMITS	The public record does not establish the specific warhead types assembled at Trekhgorny, current assembly production volumes, or the logistics protocols for HEU material transfer between Novouralsk and Trekhgorny.

DOCUMENTED STRUCTURAL CONSTRAINT Trekhgorny has no Western financial exposure. All operations are within Russia's closed nuclear weapons defense budget. OFAC SDN designation would be legally available under EO 13382 but would have no practical financial enforcement effect. Enforcement is limited to export controls on dual-use precision manufacturing equipment procurement.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
HEU input	From Novouralsk (UECC, Node 28) — approximately 120 km logistics distance
Design specifications	Component requirements received from Sarov (RFNC-VNIIEF, Node 38)
Assembly output	Russian strategic and tactical nuclear weapons inventory
Peer node	Lesnoy (Node 40, Sverdlovsk Oblast) — parallel warhead assembly facility

III. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	BIS Entity List authority restricts U.S. technology exports to listed entities regardless of SDN status [TIER 1 — EAR Part 744]. Precision manufacturing equipment — vacuum equipment, specialized machining tools, high-strength materials — relevant to warhead component assembly is controlled under various ECCN classifications.
JUDGMENT	Available reporting supports a moderate-confidence judgment that BIS Entity List designation targeting precision manufacturing equipment procurement is the operative enforcement instrument at Trekhgorny. The Novouralsk-Trekhgorny HEU logistics corridor warrants surveillance as the primary material accountability gap in the Russian weapons complex.
LIMITS	The public record does not establish which specific precision manufacturing equipment categories are being procured for Trekhgorny through third-country intermediaries, or the current operational status of the facility's assembly programs.

IV. ENFORCEMENT GAPS

Gap 1 — Novouralsk-Trekhgorny Corridor Unmonitored

The logistics corridor between UECC (Novouralsk, Node 28) and the Trekhgorny assembly facility is not subject to any independent monitoring mechanism. This is the shortest HEU-to-assembly route in the Russian weapons complex. HIGH CONFIDENCE based on the absence of any publicly documented monitoring mechanism for this corridor.

Gap 2 — No Commercial Enforcement Leverage

Trekhgorny has no Western financial exposure. Enforcement is limited to export controls on dual-use procurement. This is a structural constraint that cannot be resolved through sanctions tools alone.

V. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Development of an export controls assessment targeting precision manufacturing equipment procurement for Trekhgorny through third-country intermediaries.
2. Incorporation of Novouralsk-Trekhgorny HEU logistics corridor into fissile material accountability modeling for any future arms control verification planning.
3. Pairing of Trekhgorny and Lesnoy (Node 40) in enforcement planning as functionally complementary assembly nodes with distinct material processing capabilities.

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	FESCO named by UN Panel in DPRK-adjacent corridors but specific correspondent banks not named. Upgrade: UN Panel S/2024 annexes; OFAC enforcement releases for FESCO or affiliated vessels.

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WP-2026 SERIES

LESNOY — ELEKTROKHIMPRIBOR

Nuclear Warhead Assembly / HEU Metal Production — Closed City (Sverdlovsk-45)

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

WEAPONS TIER

CONSTRAINED

NODE ID	40 — SENI-17
LOCATION	Lesnoy, Sverdlovsk Oblast, Russia (ZATO)
COORDINATES	58.63°N, 59.79°E
OPERATOR	Elektrokhimpribor / Rosatom
NETWORK GROUP	Russian Nuclear Weapons Complex — Assembly Tier
PROCESS ROLE	Weapons — Nuclear Warhead Component Fabrication, Assembly, and HEU Metal Production
STATUS	CONSTRAINED — no Western financial exposure; HEU metal production makes this analytically distinct from Trekhgorny

KEY JUDGMENTS

KJ 1

Lesnoy hosts Elektrokhimpribor, Russia's second major warhead assembly facility. Unlike Trekhgorny, it produces uranium metal components directly from HEU feedstock — placing it at the intersection of the enrichment and assembly tiers and creating an additional material accountability requirement.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

The UECC-Lesnoy HEU material transfer corridor — approximately 200 km within Sverdlovsk Oblast — is the most significant unmonitored weapons-grade material movement in the Russian weapons complex. No independent accountability

mechanism covers this transfer.

LIKELY — MODERATE CONFIDENCE

I. FACILITY OVERVIEW

ATTRIBUTION	NTI's Russia Facilities Profile and U.S. State Department arms control compliance reports identify Lesnoy — formerly Sverdlovsk-45 — as a ZATO facility hosting Elektrokhimpribor, which conducts warhead assembly and uranium metal production [TIER 2 — NTI Russia Facilities; TIER 1 — State Department Compliance Report 2023].
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that Lesnoy's uranium metal production capability makes it analytically distinct from Trekhgorny. Receiving HEU from UECC and converting it to metallic form is a discrete material processing step with its own accountability requirement. The UECC-Lesnoy HEU flow is the primary material accountability gap in the Sverdlovsk Oblast weapons complex cluster.
LIMITS	The public record does not establish the specific HEU quantities transferred between UECC and Lesnoy annually, the warhead types assembled at the facility, or the current operational tempo of uranium metal production.

DOCUMENTED STRUCTURAL CONSTRAINT All Lesnoy operations are within Russia's closed defense budget with no Western financial exposure. Enforcement instruments are limited to export controls on uranium metallurgy equipment procurement and arms control verification mechanisms for fissile material accountability.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
HEU input	From Novouralsk (UECC, Node 28) — approximately 200 km, same Sverdlovsk Oblast
Uranium metal production	HEU → metallic uranium components for warhead assembly
Design specifications	From Sarov (RFNC-VNIIEF, Node 38)
Assembly output	Russian nuclear weapons inventory
Peer node	Trekhgorny (Node 39, Chelyabinsk Oblast) — parallel assembly facility

III. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	Fissile Material Cutoff Treaty (FMCT) negotiations and IAEA fissile material studies identify HEU metal production at weapons assembly facilities as a key accountability gap in any future verification regime [TIER 2 — IAEA STI/PUB fissile material studies; TIER 2 — NTI FMCT analysis].
JUDGMENT	Available reporting supports a moderate-confidence judgment that UECC-Lesnoy HEU material flow is the primary unmonitored weapons-grade material movement in the Russian weapons complex and should be central to any future arms control verification design for Russian HEU accountability.
LIMITS	The public record does not establish the specific quantities, frequencies, or logistics protocols for HEU transfer between UECC and Lesnoy, making precise accountability gap quantification impossible from open sources.

IV. ENFORCEMENT GAPS

Gap 1 — HEU Metal Production Accountability

Uranium metal production from HEU at Lesnoy occurs within a closed ZATO facility with no external material accountability mechanism. The quantity of HEU consumed in uranium metal production is not independently verifiable. HIGH CONFIDENCE based on the absence of any publicly documented accountability mechanism.

Gap 2 — UECC-Lesnoy Material Flow Unmonitored

The HEU transfer between Novouralsk (UECC, Node 28) and Lesnoy is the most significant unmonitored weapons-grade material movement in the Russian nuclear complex. MODERATE CONFIDENCE based on documented absence of monitoring mechanisms for this corridor.

V. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Incorporation of the UECC-Lesnoy HEU material flow into fissile material accountability analysis for any future arms control verification planning or FMCT negotiations.
2. Development of an export controls assessment targeting uranium metallurgy equipment procurement for Lesnoy through third-country intermediaries.
3. Pairing of Lesnoy and Trekhgorny profiles in enforcement planning; the two facilities have complementary capabilities and should be addressed as a cluster rather than individually.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

MAYAK PRODUCTION ASSOCIATION

Plutonium Reprocessing / Spent Fuel / Weapons-Grade Pu Stockpile

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

REPROCESSING TIER

TARGETABLE

NODE ID	41 — SENI-18
LOCATION	Ozyorsk, Chelyabinsk Oblast, Russia (ZATO)
COORDINATES	55.71°N, 60.85°E
OPERATOR	Mayak Production Association / Rosatom
NETWORK GROUP	Russian Nuclear Weapons Complex — Reprocessing / Pu Pathway
PROCESS ROLE	Weapons / Civilian — Spent Fuel Reprocessing, Weapons-Grade Pu Management, Naval Fuel Reprocessing
STATUS	TARGETABLE (commercial ops) / CONSTRAINED (Pu ops) — commercial reprocessing provides enforcement lever; PMDA suspension is the primary arms control gap

KEY JUDGMENTS

KJ 1

Mayak is Russia's only operating spent nuclear fuel reprocessing facility and the primary site for weapons-grade Pu stockpile management. The 2016 Russian suspension of the Plutonium Management and Disposition Agreement (PMDA) eliminated independent accountability for approximately 34 metric tons of weapons-grade Pu [TIER 1 — State Department PMDA Compliance Reports; EO 13883 (2019)].

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2	Mayak's commercial reprocessing operations — civilian spent fuel processing fees — generate financially visible revenue flows accessible to OFAC enforcement. The weapons Pu management operations have no Western financial exposure. **LIKELY — MODERATE CONFIDENCE**
KJ 3	Naval spent fuel reprocessed at Mayak is exempt from IAEA civilian safeguards obligations. Pu separated from naval spent fuel has no independent accountability mechanism. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on known IAEA safeguards scope limitations for naval nuclear fuel. **LIKELY — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	U.S. State Department arms control compliance reports and DOE/NNSA program documentation identify Mayak Production Association as Russia's primary reprocessing facility and the site of Russia's weapons-grade Pu stockpile management [TIER 1 — State Department Compliance Report 2023; DOE/NNSA CTR Program Reports]. Russia suspended its PMDA obligations in 2016 under Federal Law 392-FZ [TIER 1 — PMDA text and State Department compliance records].
JUDGMENT	Official reporting supports a high-confidence judgment that Mayak's PMDA suspension created the most significant unresolved weapons Pu accountability gap in the Russian nuclear complex. Restoring PMDA verification or establishing an equivalent mechanism is a higher-priority policy objective than OFAC designation of Mayak's commercial operations.
LIMITS	The public record does not establish the current quantity of weapons-grade Pu stored at Mayak, the separation volume from civilian versus naval spent fuel reprocessing, or the current status of any Pu disposition activities.

DOCUMENTED STRUCTURAL CONSTRAINT Russia's 2016 suspension of PMDA obligations eliminated the accountability framework for 34 metric tons of weapons-grade Pu. OFAC designation of Mayak's commercial operations would not address this weapons Pu accountability gap, which requires arms control and diplomatic engagement rather than sanctions enforcement.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
Spent fuel input	Receives spent fuel from Russian VVER reactors and naval nuclear fleet

Separated Pu output	Weapons-grade Pu stockpile management (legacy); civilian separated Pu → potential MOX fabrication
Civilian MOX pathway	→ VNIINM Bochvar (Moscow) — MOX fuel fabrication node (separate SENI profile recommended)
Geographic cluster	Chelyabinsk Oblast proximity: Trekhgorny (~150 km, Node 39), Lesnoy (~200 km, Node 40)

III. FINANCIAL NETWORK

ATTRIBUTION	Rosatom consolidated financial reporting identifies civilian spent fuel storage and reprocessing fees as a commercial revenue stream for Mayak. Some international spent fuel reprocessing contracts — for foreign research reactor fuel — generate financially visible flows [TIER 2 — Rosatom annual reports; TIER 3 — Nuclear Engineering International].
JUDGMENT	Available reporting supports a moderate-confidence judgment that Mayak's commercial reprocessing revenue is accessible to OFAC enforcement in ways that the weapons Pu management operations are not. This commercial lever is real but insufficient to address the PMDA accountability gap.
LIMITS	The public record does not establish the specific Western banks clearing Mayak commercial reprocessing fees, the contract values of international reprocessing agreements, or whether any Western financial institutions currently have exposure to Mayak commercial transactions.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	Mayak designatable as Pu production and management facility. Both weapons and civilian operations provide proliferation predicate.
CAATSA §231	Defense/Intel Sector	Mayak's weapons Pu management places it within Rosatom's defense programs.
IEEPA / OFAC	Commercial Reprocessing	Commercial reprocessing revenue provides partial financial enforcement leverage.
PMDA / Arms Control	Pu Disposition Agreement	PMDA restoration is the primary policy instrument for Mayak Pu accountability — not a

		sanctions tool.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC has authority to designate Mayak under EO 13382 based on its Pu production history and ongoing reprocessing operations. The PMDA was suspended by Russia under Federal Law 392-FZ in 2016; the U.S. State Department has reported this suspension annually in arms control compliance reports since 2017 [TIER 1 — State Department Annual Compliance Reports 2017–2024].
JUDGMENT	Official reporting supports a moderate-confidence judgment that OFAC designation of Mayak's commercial reprocessing entity is actionable and would generate financial pressure on the commercial revenue stream. However, the most significant enforcement gap at Mayak — the PMDA Pu accountability void — requires arms control engagement, not sanctions enforcement.
LIMITS	The public record does not establish whether OFAC designation of Mayak's commercial operations would disrupt any current Western financial institution exposure to Mayak transactions, or whether the commercial reprocessing revenue is already cleared through non-Western channels.

VI. ENFORCEMENT GAPS

Gap 1 — PMDA Suspended — Pu Accountability Framework Absent

Russia's 2016 PMDA suspension eliminated independent accountability for approximately 34 metric tons of weapons-grade Pu at Mayak. No equivalent verification mechanism has replaced it. This is the most significant weapons material accountability gap in the Russian nuclear complex. HIGH CONFIDENCE based on documented PMDA suspension and State Department compliance reporting.

Gap 2 — Naval Spent Fuel Pu Separation Unmonitored

Naval spent fuel reprocessing at Mayak is exempt from IAEA civilian safeguards. Pu separated from naval fuel has no independent accountability mechanism. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on IAEA safeguards scope limitations for naval nuclear programs.

Gap 3 — Civilian Pu Separation Volume Not Independently Verified

IAEA safeguards coverage of Mayak's civilian reprocessing provides partial Pu accountability, but adequacy of safeguards at a facility with Mayak's operational complexity and mixed civilian-military history is not confirmed. MODERATE CONFIDENCE — ANALYTICALLY INFERRED.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. A State Department action to prioritize PMDA restoration or equivalent Pu disposition verification mechanism as the primary policy objective for Mayak — this is an arms control and diplomatic priority above any sanctions enforcement action.

2. Development of a Mayak commercial reprocessing entity designation package under EO 13382 as a parallel financial pressure instrument.

3. Engagement through IAEA on enhanced safeguards for Mayak's civilian reprocessing operations, specifically addressing material accountancy for separated Pu output.

4. Development of policy analysis on naval spent fuel reprocessing safeguards exemption — assess whether a voluntary monitoring mechanism for naval Pu separation at Mayak is achievable.

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

ZHELEZNOGORSK — MINING CHEMICAL COMBINE

Underground Pu Reprocessing / Storage — Closed City (Krasnoyarsk-26)

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

REPROCESSING TIER

TARGETABLE

NODE ID	42 — SENI-19
LOCATION	Zheleznogorsk, Krasnoyarsk Krai, Russia (ZATO)
COORDINATES	56.25°N, 93.53°E
OPERATOR	Mining Chemical Combine (GKhK) / Rosatom
NETWORK GROUP	Russian Nuclear Weapons Complex — Reprocessing / Pu Pathway
PROCESS ROLE	Weapons / Civilian — Pu Reprocessing, Spent Fuel Underground Storage, RT-2 MOX Fabrication (planned)
STATUS	TARGETABLE — SDN available; RT-2 procurement chain is the primary actionable financial target; underground construction limits physical enforcement utility

KEY JUDGMENTS

KJ 1

GKhK at Zheleznogorsk is Russia's second major Pu reprocessing site and hosts the largest underground spent nuclear fuel storage complex in Russia. Its ADE-2 reactor produced weapons-grade Pu until 2010 — after PMDA signing (2000) but before reactor shutdown — creating a specific material accountability gap for final-years Pu production.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ

GKhK's underground construction — key production areas 200–300 meters below

2	surface — makes physical interdiction operationally infeasible and limits satellite imagery characterization. Financial enforcement through spent fuel storage fee revenue and RT-2 construction procurement is the operative approach. **ALMOST CERTAINLY — HIGH CONFIDENCE**
KJ 3	If the incomplete RT-2 reprocessing plant at GKhK is completed, Russia would have a second industrial-scale Pu separation facility alongside Mayak. RT-2 completion would significantly increase separated Pu output beyond any plausible civilian MOX fuel demand. MODERATE CONFIDENCE based on RT-2 capacity specifications in NTI and IAEA reporting. **LIKELY — MODERATE CONFIDENCE**

I. FACILITY OVERVIEW

ATTRIBUTION	NTI's Russia Nuclear Facilities Profile and IAEA country documentation identify the Mining Chemical Combine (GKhK) at Zheleznogorsk — formerly Krasnoyarsk-26 — as a ZATO facility with underground Pu reprocessing, spent fuel storage, and incomplete RT-2 plant construction [TIER 2 — NTI Russia Facilities 2023; TIER 1 — IAEA CNPP Russia 2024]. ADE-2 reactor operated until 2010 as a dual-use heat-and-power source [TIER 1 — DOE/NNSA CTR Program documentation].
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that GKhK is the second node in the Russian Pu pathway alongside Mayak. RT-2 completion would double Russia's Pu separation capacity, creating a strategic nonproliferation imperative to disrupt the construction procurement chain.
LIMITS	The public record does not establish RT-2's current construction status or projected completion timeline with precision. Published estimates are Tier 2–3 at best. The public record does not permit a firm conclusion on the quantity of Pu produced by ADE-2 in its final operational years (2000–2010).

DOCUMENTED STRUCTURAL CONSTRAINT GKhK's underground construction makes physical enforcement operationally infeasible. Satellite imagery provides limited facility characterization. Financial enforcement through commercial spent fuel storage revenue and RT-2 construction procurement chain interdiction are the only viable enforcement vectors.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
Spent fuel input	Receives spent nuclear fuel from Russian VVER reactors via Trans-Siberian rail

Pu storage	Legacy weapons-grade Pu from ADE-2 production; civilian separated Pu from reprocessing
RT-2 MOX pathway	Planned RT-2 → MOX fuel → BN-800/BN-1200 fast reactors (future)
Rosatom coordination	Coordinated national Pu stockpile management with Mayak (Node 41)
Regional cluster	Krasnoyarsk Krai: same region as Zelenogorsk ECP (Node 30)

III. FINANCIAL NETWORK

ATTRIBUTION	Rosatom reporting identifies civilian spent fuel storage fees as a commercial revenue stream for GKhK. RT-2 construction has consumed significant Rosatom capital expenditure; construction contractor relationships and equipment procurement have historically involved Western suppliers [TIER 2 — Rosatom annual reports; TIER 3 — Nuclear Engineering International].
JUDGMENT	Available reporting supports a moderate-confidence judgment that RT-2 construction procurement represents the primary Western-accessible financial target at GKhK. Disrupting RT-2 completion prevents Russia from establishing a second industrial-scale Pu separation facility.
LIMITS	The public record does not establish which specific construction contractors or equipment suppliers are currently providing RT-2 construction services, or the extent of Western financial exposure in the current procurement chain post-2022.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	GKhK designatable as Pu reprocessing and storage facility.
CAATSA §231	Defense/Intel Sector	ADE-2 weapons Pu production history and continued Pu management place GKhK within Rosatom's defense programs.
IEEPA / OFAC	Commercial Designation	Spent fuel storage fee revenue provides partial commercial financial leverage.
PMDA / Arms Control	ADE-2 Pu Accountability	ADE-2 final-years Pu disposition is a specific accountability gap requiring PMDA-successor

		mechanism engagement.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC has authority to designate GKhK under EO 13382 based on its Pu reprocessing history and ongoing storage operations. RT-2 construction procurement has historically involved European suppliers for reactor components and process equipment [TIER 3 — Nuclear Engineering International; NTI facilities data].
JUDGMENT	Available reporting supports a moderate-confidence judgment that RT-2 construction procurement chain targeting is the highest-leverage enforcement action at GKhK: it prevents completion of a second Pu separation facility without requiring direct action against the underground reprocessing operations that are physically inaccessible to conventional enforcement.
LIMITS	The public record does not establish the current RT-2 completion status or whether Western suppliers continue to provide construction services post-2022 under any carve-out arrangement.

VI. ENFORCEMENT GAPS

Gap 1 — ADE-2 Final-Years Pu Accountability

Pu produced by ADE-2 between PMDA signing (2000) and reactor shutdown (2010) has no independently verified disposition record. MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on ADE-2 operational parameters that the undisposed quantity may reach several hundred kilograms of weapons-grade Pu over the decade.

Gap 2 — RT-2 Completion Status Uncertain

RT-2's current construction status and projected completion timeline are not reliably documented in open-source reporting. LOW-MODERATE CONFIDENCE on any specific completion estimate. Completion would double Russia's Pu separation capacity.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Development of a GKhK SDN designation package under EO 13382, paired with Mayak designation to signal comprehensive Pu pathway enforcement.
2. BIS and A Treasury action to target the RT-2 construction procurement chain — identify equipment suppliers and engineering contractors with Western financial exposure for secondary sanctions action.
3. Engagement through IAEA on ADE-2 final-years Pu material accountability as a specific gap in the PMDA verification record.
4. A request through appropriate channels for allied intelligence assessment of RT-2 current construction status and projected completion timeline.

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	Commercial reprocessing revenue partially visible in Rosatom consolidated reports; clearing banks not named. Upgrade: IAEA safeguards reporting for commercial reprocessing; Rosatom GAAP subsidiary disclosures for international research reactor contracts.
2	DESIGNATION	Mayak commercial operations distinguishable from weapons-grade Pu ops (PMDA domain). OFAC commercial designation available under EO 13382 without triggering PMDA issues. Designation package for commercial reprocessing nexus recommended.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

DIMITROVGRAD — NIIAR

Nuclear Research Reactor Complex — Fast Reactor Fuel Research — Cf-252 Production

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

RESEARCH TIER

TARGETABLE

NODE ID	43 — SENI-20
LOCATION	Dimitrovgrad, Ulyanovsk Oblast, Russia
COORDINATES	54.22°N, 49.60°E
OPERATOR	NIIAR (Research Institute of Atomic Reactors) / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Research Tier
PROCESS ROLE	Research — Reactor Operations, Isotope Production (incl. Cf-252), Fast Reactor Fuel Qualification
STATUS	TARGETABLE — Cf-252 ECCN 1C236 controls immediately actionable; NIIAR and Isotope JSC designation available

KEY JUDGMENTS

KJ 1

NIIAR produces californium-252 (Cf-252), a neutron initiator controlled under BIS ECCN 1C236 with direct nuclear weapons trigger applications. Cf-252 export end-use monitoring from NIIAR through Isotope JSC is inadequate based on the absence of documented BIS enforcement actions against identified purchasers.

LIKELY — MODERATE CONFIDENCE

KJ 2

NIIAR's BOR-60 fast reactor program generates irradiation data used in BN-800/BN-1200 fast reactor design — reactors intended to consume separated Pu from Mayak (Node 41) and Zheleznogorsk (Node 42). This connects NIIAR analytically to the Pu

KJ 3	pathway, not only the research tier. **LIKELY — MODERATE CONFIDENCE** The extent of data sharing between NIIAR's BOR-60 fast reactor program and Sarov's weapons design programs is not documented in open-source reporting. This assessment does not establish that such sharing occurs. LOW CONFIDENCE — ANALYTICALLY INFERRED based on known institutional relationships between research and weapons design institutions in the Russian nuclear complex. **ROUGHLY EVEN ODDS — LOW CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	IAEA Nuclear Research Reactor databases and NTI Russia Nuclear Facilities Profile identify NIIAR at Dimitrovgrad as Russia's largest nuclear research center, operating six research reactors including the SM-3 high-flux reactor and BOR-60 fast neutron reactor [TIER 1 — IAEA Research Reactor Database; TIER 2 — NTI Russia Profile 2023]. BIS ECCN 1C236 controls californium-252 as a nuclear-related material.
JUDGMENT	Official reporting supports a high-confidence judgment that NIIAR's Cf-252 production is the immediately actionable enforcement target. BIS ECCN 1C236 controls provide a specific, legally grounded instrument that does not require OFAC SDN designation and can be implemented through enhanced export licensing review.
LIMITS	The public record does not establish NIIAR's annual Cf-252 production volume, the complete Isotope JSC customer list for Cf-252 sales, or end-use verification for any declared legitimate applications. The BOR-60/Sarov data sharing concern is LOW CONFIDENCE — ANALYTICALLY INFERRED and requires corroboration before publication.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
BOR-60 fast reactor	Fast reactor fuel irradiation data → BN-800/BN-1200 fast reactor design programs
Isotope production	Cf-252, Mo-99, other isotopes → Isotope JSC distribution (medical, industrial, research)
Fuel irradiation testing	TVEL JSC fuel qualification for new fuel configurations
Research coordination	Sarov (RFNC-VNIIEF, Node 38) — institutional relationship; data sharing extent LOW CONFIDENCE — ANALYTICALLY

	INFERRED

III. FINANCIAL NETWORK

ATTRIBUTION	NIIAR generates commercial revenue through radioisotope sales (via Isotope JSC), international irradiation services, and Rosatom internal research contracts. Isotope JSC's Western pharmaceutical and industrial customer relationships provide the most accessible Western financial exposure at NIIAR [TIER 2 — Isotope JSC commercial documentation; TIER 3 — industry trade press].
JUDGMENT	Available reporting supports a moderate-confidence judgment that Isotope JSC's Cf-252 and medical isotope sales represent an active Western financial flow from NIIAR operations that is not captured by any current designation. Isotope JSC designation paired with BIS ECCN 1C236 controls would address both the financial and export control dimensions simultaneously.
LIMITS	The public record does not establish the specific Western customers for NIIAR-sourced Cf-252 via Isotope JSC, the contract values, or the clearing banks. Post-2022 migration of some isotope transactions to non-Western clearing is likely but not confirmed.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EAR / BIS	ECCN 1C236 (Cf-252)	Cf-252 export controls. Immediately actionable. Does not require OFAC SDN designation.
EO 13382	WMD Proliferators	Cf-252 production and fast reactor weapons-relevant research provide independent WMD predicate.
IEEPA / OFAC	Isotope JSC Designation	Isotope JSC not currently designated. Separate action required.
CAATSA §231	Defense/Intel Sector	Fast reactor weapons fuel research and Sarov coordination (ANALYTICALLY INFERRED) place NIIAR within defense-adjacent Rosatom programs.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	BIS ECCN 1C236 controls Cf-252 as a nuclear-related material requiring export licensing. OFAC has authority under EO 13382 to designate entities involved in WMD proliferation-relevant activities [TIER 1 — BIS CCL ECCN 1C236; OFAC EO 13382].
JUDGMENT	Official authority supports a high-confidence judgment that BIS enhanced end-use monitoring and license review for all Cf-252 exports from NIIAR through Isotope JSC is the highest-priority immediately actionable enforcement step at this node. NIIAR and Isotope JSC designation under EO 13382 should follow as the second-tier action.
LIMITS	The public record does not permit a firm conclusion on whether any current Cf-252 exports from NIIAR/Isotope JSC reach DPRK-adjacent or other proliferation-relevant purchasers.

VI. ENFORCEMENT GAPS

Gap 1 — Cf-252 End-Use Monitoring Inadequate

Cf-252 exports from NIIAR through Isotope JSC are not comprehensively tracked for end-use. The complete customer list and end-use verification are not documented in open-source analysis. MODERATE CONFIDENCE based on the absence of documented BIS enforcement actions against identified purchasers.

Gap 2 — BOR-60 / Sarov Data Sharing

The extent of fast reactor fuel physics data sharing between NIIAR's BOR-60 program and Sarov's weapons design programs is not documented in open-source reporting. LOW CONFIDENCE — ANALYTICALLY INFERRED. This assertion requires corroboration before publication.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Implementation of enhanced end-use monitoring and license review for all Cf-252 exports from NIIAR/Isotope JSC under ECCN 1C236 — immediately actionable without OFAC designation.
2. Development of NIIAR and Isotope JSC designation packages under EO 13382 based on Cf-252 production and fast reactor weapons-relevant research.
3. An OFAC action to pair NIIAR and Isotope JSC designation as a consolidated action covering both the production facility and the commercial distribution entity.
4. A coordinated State-BIS effort to assess the Cf-252 export customer list through available trade and customs data; flag any customers with DPRK, Iran, or proliferation-adjacent network connections.

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	RT-2 MOX plant procurement contractors not identified in OSINT. Upgrade: EU export control licensing data for nuclear equipment

		exports to Russia; IAEA safeguards for plutonium accountability at GKhK.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

MOSCOW — KURCHATOV INSTITUTE

Nuclear Physics Research — Weapons Design Nexus — Academic-Industrial Interface

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

RESEARCH TIER

CONSTRAINED

NODE ID	44 — SENI-21
LOCATION	Moscow, Russia
COORDINATES	55.80°N, 37.40°E
OPERATOR	NRC Kurchatov Institute / Rosatom (affiliated) / Russian Government
NETWORK GROUP	Russian Nuclear Weapons Complex — Research / Design Nexus
PROCESS ROLE	Research — Nuclear Physics, Reactor Design, Weapons-Adjacent Science, International Scientific Interface
STATUS	CONSTRAINED — BIS Entity List is more appropriate than OFAC SDN; political cost from scientific community warrants be weighed

KEY JUDGMENTS

KJ 1 The Kurchatov Institute is Russia's founding nuclear research institution and the intellectual nexus connecting the academic nuclear physics community to the weapons design programs at Sarov (Node 38). Its international scientific engagement creates a technology transfer channel that does not exist at closed-city weapons facilities.

LIKELY — MODERATE CONFIDENCE

KJ 2 BIS Entity List designation targeting Western scientific equipment and software reaching Kurchatov through third-country procurement channels is more appropriate

KJ 3	and operationally targeted than OFAC SDN designation, which would have limited financial impact given Kurchatov's primarily domestic budget. **LIKELY — MODERATE CONFIDENCE** The extent of researcher movement between Kurchatov and Sarov, and the classification status of Kurchatov's nuclear data libraries, are not documented in open-source reporting. The depth of the weapons design linkage is the primary analytical uncertainty determining enforcement priority at this node. LOW CONFIDENCE — ANALYTICALLY INFERRED. **ROUGHLY EVEN ODDS — LOW CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	ODNI reporting and State Department arms control compliance documentation identify the Kurchatov Institute as Russia's primary nuclear physics research institution with historical involvement in the Soviet weapons program [TIER 1 — ODNI ATA 2024; State Department Compliance Report 2023]. The Institute has participated in international programs including ITER fusion research (participation suspended 2022) and has maintained publication relationships with Western scientific journals.
JUDGMENT	Official reporting supports a moderate-confidence judgment that Kurchatov's international scientific engagement creates a dual-use knowledge transfer channel distinct from the closed-city weapons facilities. The enforcement objective at Kurchatov is technology transfer restriction — preventing Western dual-use scientific knowledge from flowing into the weapons design complex — not financial disruption.
LIMITS	The public record does not establish the current scope of Kurchatov's remaining Western scientific relationships post-2022, the classification status of its nuclear data libraries, or the specific nature of its institutional relationship with Sarov's weapons design programs.

DOCUMENTED STRUCTURAL CONSTRAINT Kurchatov's designation as a scientific research institution with broad international academic relationships would generate significant pushback from the Western scientific community. This political cost is real and warrants be weighed against enforcement benefit. BIS Entity List designation is less politically charged than OFAC SDN and better calibrated to the technology transfer objective at this node.

II. NETWORK CONNECTIONS

CONNECTION	DETAIL
Scientific personnel	Exchange with Sarov (RFNC-VNIIEF, Node 38) — weapons

	design linkage; depth LOW CONFIDENCE — ANALYTICALLY INFERRED
Nuclear data libraries	→ Sarov weapons design calculations, TVEL fuel cycle modeling — classification status uncertain
Research reactor operations	Fuel qualification testing complementary to NIIAR (Node 43)
International engagement	ITER (suspended 2022); residual scientific exchange through publication networks and third-country channels

III. FINANCIAL NETWORK

ATTRIBUTION	The Kurchatov Institute receives Russian federal budget and Rosatom program funding. It historically received international scientific cooperation funding from Western sources including EU Horizon program grants and ITER program contributions. Post-2022 direct Western funding has been largely suspended [TIER 2 — EU Horizon program documentation; ITER Organization reports].
JUDGMENT	Available reporting supports a low-moderate-confidence judgment that residual Western financial relationships through scientific publishing, conference participation, and third-country research partnerships remain at Kurchatov post-2022 but at reduced scale. These provide modest but real financial enforcement leverage.
LIMITS	The public record does not establish the current value of Kurchatov's Western-sourced funding post-2022, the specific third-country research partnerships maintaining financial relationships with Western institutions, or whether any current transactions clear through Western correspondent banks.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EAR / BIS	Entity List	Primary instrument. Restricts U.S. technology, software, scientific equipment exports to Kurchatov.
EO 13382	WMD Proliferators	Weapons design nexus and Sarov personnel exchange provide proliferation predicate. Political cost from scientific community is significant.
CAATSA §231	Defense/Intel Sector	Weapons-adjacent research and Rosatom

		affiliation place Kurchatov within scope.
Visa / Travel Controls	Scientific Exchange Restrictions	Restricting Russian nuclear scientists' access to Western conferences. Available administratively without formal designation.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	BIS Entity List authority restricts U.S. technology exports to listed entities without requiring OFAC SDN designation [TIER 1 — EAR Part 744]. Visa and travel controls on scientists from designated institutions are implementable through State Department administrative procedures.
JUDGMENT	Available authority supports a moderate-confidence judgment that the combination of BIS Entity List designation and targeted visa controls on Kurchatov scientists with documented weapons-adjacent research programs is the most appropriate and proportionate enforcement package at this node.
LIMITS	The public record does not establish the specific scientific equipment categories being procured by Kurchatov through third-country channels, or the current travel patterns of Kurchatov scientists to Western institutions post-2022.

VI. ENFORCEMENT GAPS

Gap 1 — Personnel Exchange with Sarov Undocumented

The extent of researcher movement between Kurchatov and Sarov is not documented in open-source reporting. Understanding the depth of this linkage is the primary analytical prerequisite for calibrating enforcement priority at this node. LOW CONFIDENCE — ANALYTICALLY INFERRED.

Gap 2 — Residual Western Relationships Post-2022

The full scope of Kurchatov's remaining Western scientific relationships — through third-country intermediaries, publication networks, and informal channels — has not been comprehensively assessed since 2022.

Gap 3 — Nuclear Data Library Classification Status

The classification status of Kurchatov's nuclear data libraries and the extent to which they are shared with Sarov or third-state programs is not determinable from open-source information. LOW CONFIDENCE — ANALYTICALLY INFERRED based on known institutional practices in the Russian nuclear complex.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Development of a Kurchatov Institute Entity List designation package as the primary enforcement instrument — technology transfer restriction rather than financial blocking.

2. Implementation of targeted visa and travel controls on Kurchatov scientists with documented weapons-adjacent research programs.
3. IC-connected analysis to map residual Western scientific relationships through publication co-authorship networks, conference participation records, and third-country research partnerships.
4. Assessment of the nuclear data library sharing relationship between Kurchatov and Sarov through allied technical intelligence channels — the depth of this connection determines the enforcement priority of the node.

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	FPA-01 v3 achieves D2=[4+]: Wells Fargo named as Lantheus banking institution via SEC S-1; clearing pathway inferred not transaction-confirmed. Upgrade: NRC ADAMS 10 CFR 110.50 pre-shipment notifications for Isotope JSC Cf-252 imports would confirm clearing bank. Lantheus 10-K credit facility section would name transaction banking relationships.

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